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Low-Carbon Development Strategies for Power Generation Expansion in Sub-Saharan Africa: Insights from an Optimisation-Based Analysis for Kenya

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Abstract: Energy production and consumption are major contributors to global anthropogenic greenhouse gas emissions. Sub-Saharan African countries face the challenge of harnessing diverse energy sources to meet rising demand affordably while curbing emissions. This study uses the optimisation-based Kenya-TIMES model to explore low-carbon strategies for Kenya's power generation from 2020 to 2050. A business-as-usual (BAU) scenario is compared with four low-carbon scenarios: carbon tax, renewable portfolio standard, renewable energy subsidies, and a hybrid of subsidies and carbon tax. The analysis reveals that geothermal, wind, and hydropower dominate the energy mix until 2035 across all scenarios. After 2035, coal capacity in the BAU scenario is replaced by solar, gas, and biomass in low-carbon scenarios. While all low-carbon strategies, except the renewable energy subsidy scenario, meet Kenya's nationally determined contribution (NDC) emission reduction targets by 2050, the hybrid scenario emerges as the most effective and cost-efficient pathway. Although achieving significant emissions reductions, the carbon tax and renewable portfolio standard scenarios result in higher system costs. The results indicate that an integrated optimisation-based approach can identify optimal energy development pathways that leverage local resources to accommodate growth and enhance energy access while minimising costs and emissions.

Keywords: Sub-Saharan Africa; Kenya; low-carbon scenario; power generation; policy instruments; nationally determined contribution; greenhouse gas emissions

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1. Introduction

Globally, energy production and consumption represent by far the largest source of greenhouse gas (GHG) emissions from anthropogenic activities [1], with the burning of coal, natural gas, and oil for electricity and heat accounting for 25% of total emissions [2]. Governments in most high-income countries have aligned their energy strategies to achieve low-carbon power development pathways using policy instruments, such as environmental taxes, subsidies, feed-in tariffs, and command and control policies, such as renewable portfolio standard (RPS) and energy efficiency standards [3]. Setting up such strategies has proven difficult for low-income countries such as those found in Sub-Saharan Africa (SSA), where there is a predicted significant growth in energy demand due to projected population growth [4], economic development, and accelerated urbanisation [5]. The SSA governments face the challenge of providing affordable energy for their respective population's current demand–supply deficit and future growing demand while limiting GHG emissions and costs [6–8]. Adopting and implementing efficient and effective strategies to cut energy-related GHG emissions while maintaining affordability and satisfying the energy needs in the SSA countries is critical for future global energy sustainability [9].

The energy system's comprehensiveness and complexity require advanced energy planning tools to evaluate the effectiveness of policy instruments for achieving national energy-related GHG emission reduction objectives [10,11]. The feasibility of low-carbon power demand–supply expansion strategies can be assessed by models developed using bottom-up energy optimisation planning tools such as [12–18]. Here, the bottom-up optimisation models are prioritised over the top-down simulation models because of their high level of detail [19] and ability to derive optimal solutions in terms of multiple objectives and constraints [20]. Contrary to the high-income economies, the SSA countries have not sufficiently utilised energy planning tools to inform their national low-carbon power development strategies [21]. The energy decisions are often political, and energy models are either not used, flawed, or non-transparent. Furthermore, some countries, e.g., Kenya, use limited econometric models for national energy decision-making processes [12,15,22]. In other instances, the scenario-based models used in decision-making processes rely more on assumptions and political targets than on data-driven approaches [23]. The never-ending energy poverty manifests in failing to use advanced energy planning tools to achieve efficient planning and guide energy decisions. Thus, adopting national-scale, data-driven, and advanced bottom-up energy planning models, with assumptions reflecting each country's local conditions, could partly address the SSA countries' energy poverty [21].

This paper uses an optimisation-based bottom-up energy planning model to explore the low-carbon development strategies for power generation expansion in the SSA region through a case study of Kenya. As one of the fastest-growing economies in the SSA region, Kenya's energy demand is projected to increase by 650% to 91,400 GWh in 2045 [15]. Kenya's grid-connected power is predominantly renewable, at 78%, including hydro, geothermal, wind, solar, and bioenergy, while thermal capacity accounts for 22%, from heavy fuel oil, gasoil, and kerosene. The captive and off-grid installed capacity in 2023 was 10% of the total capacity and consisted of solar, geothermal, hydro, waste-to-heat recovery, bioenergy, wind, and diesel generation. Overall, electricity access from grid and off-grid supply was 76% in 2023 [24–26]. Despite previous plans for 100% renewables by 2022 [22], Kenya did not reach that goal and is now balancing renewable sources with thermal plants and has started importing power from Ethiopia in 2023 with a 25-year contract. Through its NDC objectives, Kenya has pledged to cut GHG emissions by at least 27% of projected 2030 baseline emissions [27,28]. Strategies like switching off oil-based thermal plants, importing hydroelectric power, and encouraging renewable uptake through policies like the feed-in tariff are in place [29–31]. However, frequent strategy amendments that are decided without proper use of simulation and optimisation planning tools challenge these goals. Advanced long-term planning is essential for achieving a sustainable, affordable, low-emission electricity supply as per Kenya's NDC and Sustainable Development Goal 7 [32].

Longa and van der Zwaan used the TIAM-ECN to analyse Kenya's GHG mitigation ambitions from an entire energy system perspective [32]. They evaluated mitigation measures for energy-related emissions from transport, agriculture, industry, and power sectors and analysed scenarios with carbon caps and a carbon tax. Carvallo et al. used Solar and Wind energy Integrated with Transmission and Conventional sources (SWITCH-Kenya) to explore sustainable growth paths for power systems in emerging economies through a case study of Kenya's power system [12]. The study evaluated the impact of a carbon tax and a zero-emission policy by 2030 through a carbon cap. Wambui et al. used the LEAP modelling platform to analyse the high penetration of renewable energy resources with an energy storage system and the internalising of environmental externalities [33].

However, the mentioned studies only evaluate the impact of two policy instruments—carbon cap and carbon tax—on power demand–supply expansion. Longa and van der Zwaan [32] assess the carbon cap and carbon tax policy instruments on the entire energy system, and the model does not provide a detailed evaluation of the impact of these policy instruments on the power sector, which is projected to have the highest emissions in 2050, as presented in Luna et al. [27]. Further, excluding domestic coal supply denies the model the choice of a relatively cheap energy source. This exclusion might result in the model prioritising a less polluting fuel source with underestimated GHG emissions, contrary to what it would have selected if the more affordable domestic coal supply option were present. Lastly, the power demand forecast used in the studies is an estimate that significantly varies from the government's forecasted power demand. Wambui et al. [33] only analyse the implication of energy storage on the high-level penetration of renewables and investigate the socio-techno-economic impacts associated with energy saving and reducing greenhouse gas emissions. There is no detailed analysis of policy instruments and their implication on the energy mix, GHG emissions, and energy costs. Agutu et al. [34] discuss the importance of accounting for finance in electrification models for SSA and show that incorrectly estimated factors often affect outcomes of policy evaluation.

Despite the growing need for data-driven energy planning in SSA, studies on policy portfolios remain limited. Quayson et al. [35] and Daggash and Mac Dowell [36] discuss the role of optimisation models in assessing carbon taxes and subsidies but do not address Kenya's energy mix or subsidy-driven renewable implementation. In a similar manner, Ankel-Peters et al. [37] and IMF [38] emphasise the challenges of carbon pricing and energy subsidy reforms in SSA and underscore the need for tailored policy strategies.

The authors' review of SSA's existing national energy system models [21], as well as a more recent review by Akpahout et al. [10], indicates that using advanced energy planning tools is vital in informing effective energy decisions to achieve universal energy access while limiting GHG emissions by countries in the SSA region. It was recommended that the SSA countries should develop data-driven national-scale energy optimisation models. As a result, a national bottom-up energy system optimisation model for Kenya, the Kenya-TIMES [15], was developed using The Integrated MARKAL-EFOM System–VEDA (TIMES–VEDA) framework. The current study aims to assess the implication of three low-carbon policy instruments on Kenya's power generation expansion using a vastly improved version of the Kenya-TIMES model. The improvements include using time slices for seasons and different times of the day, which is essential for correctly capturing the characteristics of intermittent renewable energy sources and market fluctuations in demand. The new version includes the peaking equation, accounting for technology learning curves, the municipal solid waste primary energy supply option, and imported/exported power from/to neighbouring countries (see further description in Section 3.1). The new model is used in this study to evaluate the effectiveness of different policy instruments, namely carbon tax, renewable portfolio standard, and renewable subsidies, on Kenya's power generation expansion regarding the power generation technology mix, GHG emissions, and the energy system costs, as well as to inform the nation's low-carbon development strategy. The exogenous demand projection for the government's Vision 2030 blueprint demand level [15] is used, and the analysis covers the 2020 to 2050 period.

Apart from introducing an advanced version of the Kenya-TIMES model, the novelty of this study lies in the fact that it is the first to evaluate and compare the three policy instruments using one energy model and unified assumptions. The study evaluates up to the year 2050 and uses the government's most current forecasted electricity demand growth for Kenya and the energy sector's most recent data and future assumptions. Furthermore, it is the first to analyse the impact of an energy subsidy, mainly targeting solar and wind, on Kenya's power sector and the first to run a hybrid of the renewable energy subsidy and the carbon tax scenarios for Kenya's power sector, where the revenue accrued from the carbon tax is used as a subsidy for renewable uptake. Performing the analysis using a comprehensive optimisation-based modelling platform, allowing for the identification of non-trivial solutions using the low-carbon policy instruments, even further adds to the study's novelty and can result in far superior results than those created without optimisation. In the rest of the paper, Section 2 explains Kenya's current GHG emissions, Section 3 presents the methodology, and Section 4 presents results, while the discussion and conclusion are presented in Sections 5 and 6, respectively.

2. Kenya's Energy Sector Emissions and Reduction Targets

To join in the global effort to mitigate GHG emissions and their resultant impacts, the Government of Kenya ratified the United Nations Framework Convention on Climate Change (UNFCCC) in 1994 [39]. Kenya's Second National Communication (SNC) to the UNFCCC presents an inventory of national GHG emissions from 2010 to 2050 [27]. It projects the total emissions to increase by 286% from 2010 to 2050, as seen in Figure 1.

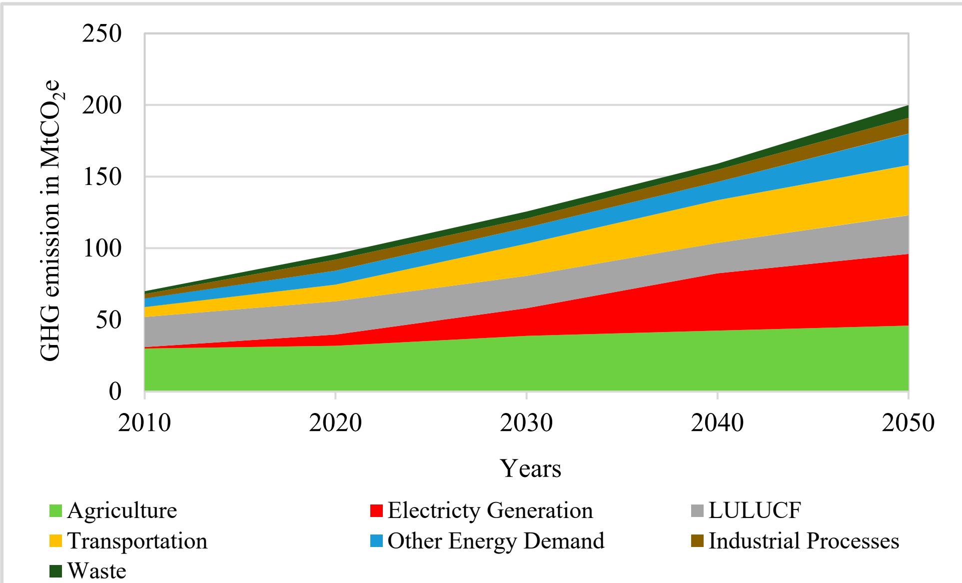

Figure 1. Sectoral GHG emission projections for seven main emission sectors in Kenya (data were derived from [27] and analysed by the authors). Predictions show an increase of 286% from 2010 to 2050, with emissions from electricity generation growing the most.

The electricity sector's emissions growth has the highest growth (714%) among all sectors. The growth is mainly due to the use of fossil fuels, driven by the rapid economic and population growth envisaged under the Vision 2030 blueprint [32]. It is critical to note that there is high uncertainty in the emission projections, especially concerning the development of emission-intensive industries [40].

Kenya has committed to cutting its overall emissions by 30% in 2030 relative to the projected 2030 baseline emissions shown in Figure 2. All seven sectors have set a reduction target, and the agencies and ministries responsible for each sector must support and implement the sector's emission cuts. Sectoral reduction potentials are derived using the Intergovernmental Panel on Climate Change (IPCC) guidelines [27]. The NDC strategy proposes a range from low to high targets depending on the level of implementation of abatement policies. For the electricity generation sector, the targets for 2030 are a 27–48% reduction compared with baseline predictions. Currently, the Kenyan government does not have emission reduction targets beyond 2030. The 2030 emission reduction target from power sector-related emissions is 30%. Accordingly, we assume the reduction target is 30% for all the periods from 2030 to 2050 and use it as a reference in the current study. It is critical to note that no empirical reference is available to select the percentage reduction target beyond 2030.

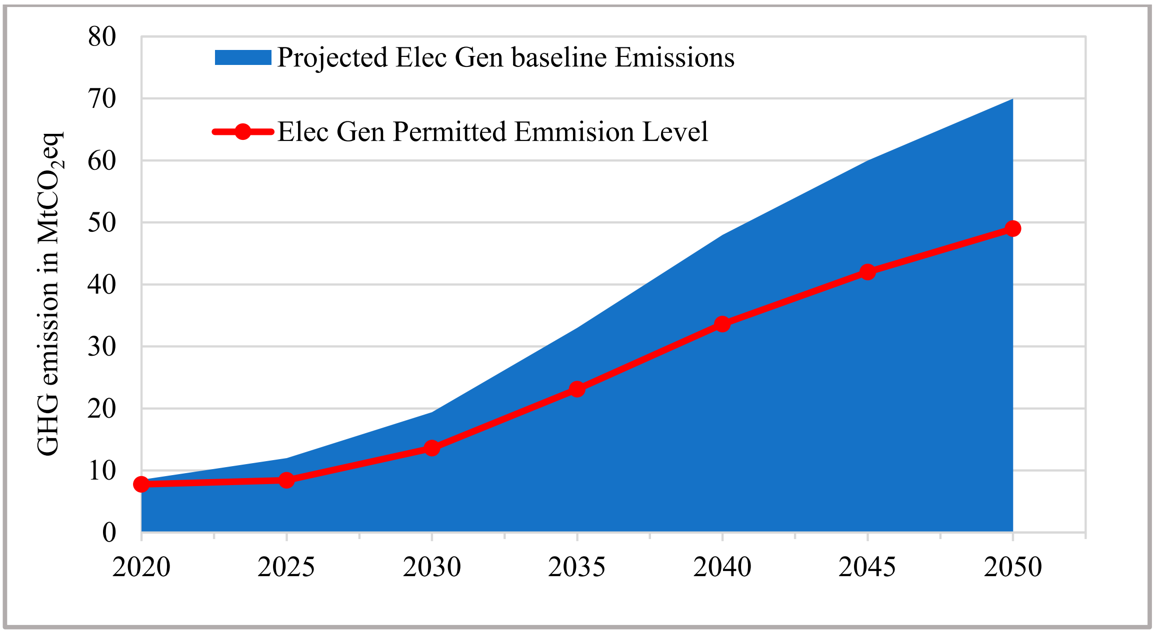

Figure 2. The projected GHG emissions from the electricity sector and a line depicting the assumed maximum permitted emission level based on a 30% reduction target.

3. Methodology

The study develops and applies a quantitative energy system planning tool, the Kenya-TIMES model, fed by actual data collected from the energy sector in Kenya, reviews of the literature, and expert predictions where actual data are unavailable. The Kenya-TIMES model is a single region integrated energy–environment–economy system optimisation model developed in the TIMES modelling framework. The methodology uses scenario analysis to appraise low-carbon policy instruments and Kenya's future demand–supply power generation expansion. The study uses the model to study how the low-carbon policy instruments affect the evolution of the installed capacity and power generation technology mix, the GHG emissions, the total costs of the energy system, the capital and operations costs, and the marginal electricity prices.

3.1. Modelling Approach

Kenya-TIMES is built in the TIMES partial equilibrium energy modelling framework, a technology-rich, bottom-up model generator that uses linear programming to produce a least-cost energy system over a medium- to long-term time horizon [41,42]. The TIMES model is based on an objective function and constraints, expressed using parameters and decision variables. Parameters are exogenous inputs, including technology costs, efficiency, availability factors, end-use energy demand, primary energy supply curves, general discounting factors, and policy instruments. Decision variables are endogenously generated model outputs. The main decision variables are new technology capacity and technology activity level. From these main decision variables, the model generates the total costs of the system discounted to net present value, marginal energy prices, and GHG emissions decision variables. All parameters and indices are written in lowercase letters in the following model equations, while all variables and sets are in uppercase (capital) letters. The indices and sets used in the model are:

y for any year in the modelling period such that $y \in Y$, where Y is the set of all the years in the modelling period
t for any given five-year time period such that $t \in T$
s for any given time slice such that $s \in S$
p for any technology or process that consumes or generates energy, e.g., demand device or energy generation technologies, such that $p \in P$
c for any commodity, e.g., energy carriers such as coal, such that $c \in C$
d for end-use energy service demand, e.g., lighting, cooking, and heating, such that $d \in D$
i for the form of energy, e.g., primary energy (energy from coal, diesel, etc.), such that $i \in I$
x for any pollutant resulting from the construction, operation, and maintenance of a power plant such that $x \in X$

Further, since Kenya's inter-day electricity demand variation is insignificant, each running year was divided into four physical quarters called seasons. A typical 24 h demand day in each of the four quarters was selected, and its demand was divided into the following time slices: Day (D) from 7 am to 6 pm, Night (N) from 10 pm to 7 am, and Peak (P) from 6 pm to 10 pm. Consequently, twelve annual time slices are considered, as illustrated in Figure 3. A twelve-year time slice approach presents a detailed accounting of the day, night, and peak capacity factors for wind and solar technologies, hence providing an efficient representation of the utilisation level of these intermittent sources in Kenya's power generation. Figure 4 presents the average daily load curve and average daily solar radiation, showing the variation between the maximum solar radiation and peak demand within a day.

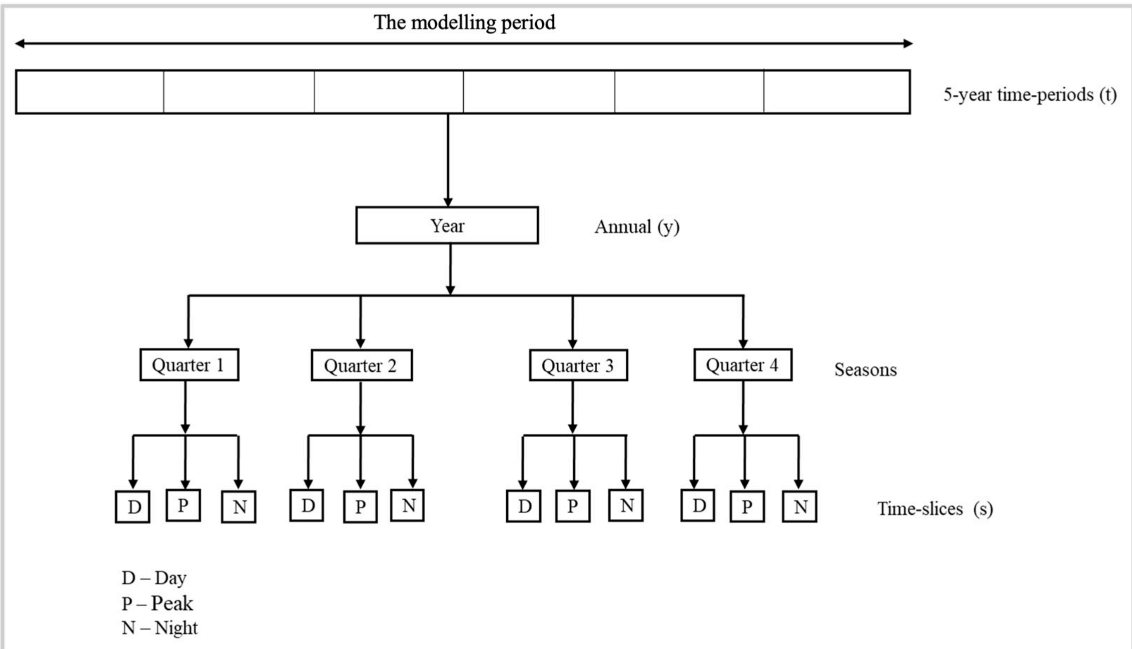

Figure 3. Time slice division for the Kenya-TIMES model. Three time slices are used to capture daily variations in demand and generation with Day (D) from 7 a.m. to 6 p.m., Night (N) from 10 p.m. to 7 a.m., and Peak (P) from 6 p.m. to 10 p.m. Four time slices are used for seasonal variations, resulting in overall twelve time slices considered each year.

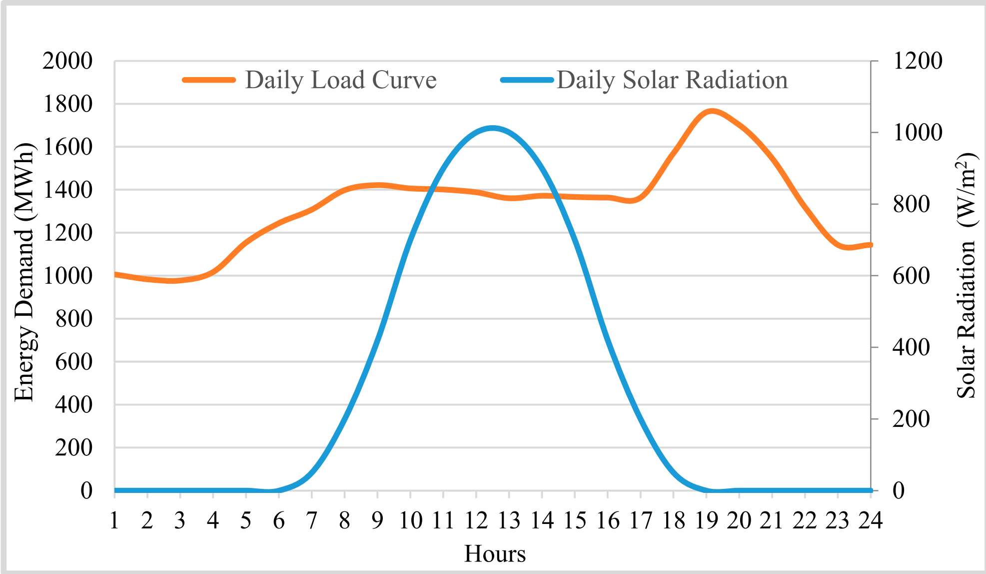

Figure 4. Kenya's average daily solar radiation profile and average daily load curve (data for solar radiation adapted from [43]).

3.1.1. The Objective Function

The objective function minimises the total costs of the whole system over the entire planning period, with all the cost elements discounted to the base year by using Equation (1). The system costs include primary energy delivery, energy transformations, operation and maintenance, and transmission and distribution costs.

$$\begin{aligned}\text{MIN OBJ} = \textstyle\sum_{y \in Y} [&DISC(y) \cdot [INVCOST(y) + FIXCOST(y) + VARCOST(y) \\ &+ELASTCOST(y) + IMPCOST(y) + EMCOST(y) - EXPREV(y) \\ &-SALVAGE]]\end{aligned} \tag{1}$$

where OBJ is the total system costs, covering all the years in the modelling horizon and discounted to the base year; $DISC(y)$ is the discounting factor for each year y; $INVCOST(y)$ is the investment costs and the associated tax and subsidies, as well as the decommissioning capital costs; $FIXCOST(y)$ is the fixed annual operational and maintenance costs; $VARCOST(y)$ is the variable operational and maintenance costs; $ELASTCOST(y)$ is the costs incurred as a result of a demand reduction; $IMPCOST(y)$ is the costs of importing a primary energy supply; $EMCOST(y)$ is the emission costs per ton of emissions if an emission tax is used; $EXPREV(y)$ is the export revenue earned from exporting energy; and $SALVAGE$ is the residual monetary value of all the investments made in the modelling period that remains at the end of the modelling horizon [42]. Subsidy values are subtracted from the costs in cases where they are applied.

3.1.2. Constraints

Constraints are defined by several equations that must be satisfied by the model when minimising the total system costs. The main constraint formulations were adopted from [13,15,42,44].

Demand satisfaction

For the total end-use energy service demand d, including transmission and distribution losses, at any given time slice s, the total available activity of technologies p providing energy for the end-use energy service demand d must satisfy the demand as calculated with Equation (2):

$$\sum_{p \in P} ACT_{p,s} \geq dem_{d,s} \quad \forall\, s \in \text{S},\ d \in \text{D} \tag{2}$$

where $dem_{d,s}$ is the (exogenously provided) gross total for end-use energy service demand d at time slice s that should be met by energy from the sum of activity level $ACT_{p,s}$ of technology p at time slice s.

Capacity utilisation

For each technology p, activity (e.g., power generation) should not exceed the installed capacity at any time slice s:

$$ACT_{p,s} \le cf_{p,s} \cdot CAP_{p,s} \; \forall \; p \in \mathrm{P}, s \in \mathrm{S} \tag{3}$$

where $cf_{p,s}$ is the capacity factor (exogenously provided) of technology p at time slice s, and $CAP_{p,s}$ is the total installed capacity.

Capacity transfer

For a given technology p, the total available capacity at time period t must be equal to the sum of capacity investments made by the model in the past and current time periods, and whose technical life has not yet ended, plus the capacity in place at the base year that is still available.

$$CAP_{p,t} = \sum_{t=0}^{t} NCAP_{p,t} + residCAP \quad \forall \; p \in \mathrm{P}, t \in \mathrm{T} \tag{4}$$

where $NCAP_{p,t}$ is the new capacity additions made by the model for technology p in the past and current time periods t; $residCAP_{p,t}$ is the exogenously provided capacity of technology p due to investments that were made before the model period and still exist at time period t.

Commodity balance equation

For a consistent flow of each energy form, consumption (including exports) must not exceed availability, which includes domestic production and imports.

$$\sum\nolimits_{p \in P} inp_{p,i} \cdot ACT_{p,s} + EXP_{i,s} \le \sum\nolimits_{p \in P} out_{p,i} \cdot ACT_{p,s} + IMP_{i,s} \quad \forall \; i \in \mathrm{I}, \; s \in \mathrm{S} \tag{5}$$

where i is the form of energy in the model; $inp_{p,i}$ is the amount of energy form i consumed by one unit of activity $ACT_{p,s}$ (e.g., petajoules of coal consumed to produce 1 MWh) of technology p at time slice s; $EXP_{i,s}$ is the amount of energy form i exported; $out_{p,i}$ is the amount of energy form i produced by one unit of activity $ACT_{p,s}$ (e.g., petajoules of coal production per hour from the mining process) of technology p; $IMP_{i,s}$ is the imported energy.

Emission constraints

An emission cap or tax can define emission-related constraints. When applied as an emission cap, the emission cap is defined as the overall emissions of certain pollutants in a given year or time period or cumulative over the whole planning horizon. The emission cap limit is expressed with Equation (6):

$$ENV_{x,y} \le env_{limt\,x,y} \quad \forall \; x \in \mathrm{X}, y \in \mathrm{Y} \tag{6}$$

where $env\_limt_{x,y}$ is the upper limit on total emissions of pollutant x in year y, and $ENV_{x,y}$ is the total emissions of pollutant x in year y, as calculated with Equation (7):

$$ENV_{x,y} = \sum\nolimits_{p \in P} \left[emINV_{x,p,y} \cdot INV_{p,y} + emCAP_{x,p,y} \cdot CAP_{p,y} + emACT_{x,p,y} \cdot ACT_{p,y} \right] \forall \; x \in X, y \in Y \tag{7}$$

where $emINV_{x,p,y}$ is the emission coefficient of pollutant x corresponding to the construction of the new investment $INV_{p,y}$ in technology p in year y; $emCAP_{x,p,y}$ is the emission coefficient linked to the unit capacity $CAP_{p,y}$ of technology p; $emACT_{x,p,y}$ is the emission coefficient of pollutant x in year y, related to the activity $ACT_{p,y}$ of technology p. An emission tax can also be formulated by Equation (8) and added as a cost element to the objective function (Equation (1)):

$$\sum_{y\in Y}\sum_{x\in X} ENV_{x,y}\cdot Etax_{x,y} \tag{8}$$

where $Etax_{x,t}$ is the emission tax coefficient (exogenously provided) on pollutant x in year y.

Other constraints

Other constraints, such as discrete investment, capacity growth rate, renewable portfolio standard, renewable energy subsidies, and peaking reserve, can be included in the model depending on the applied energy policies. A discrete investment constraint is an equality constraint that restricts investments in chosen technologies to specific discrete sizes. Capacity growth rate specifies that the market share of a particular technology or group of technologies cannot exceed a defined fraction. A constraint could also be added to limit the available investment budget and budget for incentives, but this is not applied in the scenarios studied in this work.

The renewable portfolio standard specifies that a defined minimum percentage share of electricity supply must come from designated renewable energy technologies at a stipulated time period. The subsidy constraint specifies that for every new investment in a specified renewable technology, the unit investment costs must be reduced by a percentage in accordance with intended energy policies.

The peaking reserve constraint specifies that the total available capacity of all technologies producing a commodity (e.g., electricity) at each time slice must, by a certain percentage, exceed the average demand for the commodity in the time slice where peaking occurs. The peaking reserve constraint is expressed with Equation (9):

$$\sum_{p\in P} CAP_{p,c,s}\cdot CAP\_PKCNT_{p,c,s} \geq dem_{c,s} + PKRSV_{c,s}\cdot dem_{c,s} \quad \forall\, s\in \mathrm{S}, c\in \mathrm{CG} \tag{9}$$

where $CAP_{p,s}$ is the installed capacity of technology p, producing commodity c (in this case, electricity) during peak demand time slice s; $CAP\_PKCNT_{p,s}$ specifies the fraction of technology p's capacity that contributes to the peak load for commodity c in time slice s; $dem_{c,s}$ is the peak demand capacity for commodity c produced by technology p at time slice s; $PKRSV_{c,s}$ is the percentage peak reserve factor for commodity c at time slice s.

3.2. The Kenya-TIMES Reference Energy System

The main components of the reference energy system include primary energy supply, conversion technologies, and the energy demand sector, as can be seen in Figure 5.

3.2.1. Primary Energy Supply

The energy supply in Kenya-TIMES includes fossil fuels and primary renewable energy from domestic and imported sources. The fossil fuels considered are based on the government's future planned power generation technologies [22]. They are supplied in the model from either domestic or domestic and imported processes (see Figure 5). The future prices for fossil fuels and nuclear resources are calculated based on the base year price presented in the government's Least-Cost Power Development Plan (LCPDP) [45] and the five-year global percentage increase in price as presented in the International Energy Agency's power generation technology database [46]. New to this model is biomass, which was informed by government plans for a 40 MW waste-to-energy plant in Nairobi [47]. Domestic renewable energy potential is also assessed (see Table 1), with a KenGen-funded study highlighting a 115 MW potential from Nairobi's waste by 2035, assuming a 90% waste collection rate [48]. The countrywide annual biomass potential is 3500 MW, and natural gas is 4000 MW. Additionally, Kenya plans to import 400 MW of hydroelectric power from Ethiopia's Grand Renaissance Dam hydroelectric plant [22], enhancing the model's imported electricity options with no set import limits for primary energy resources.

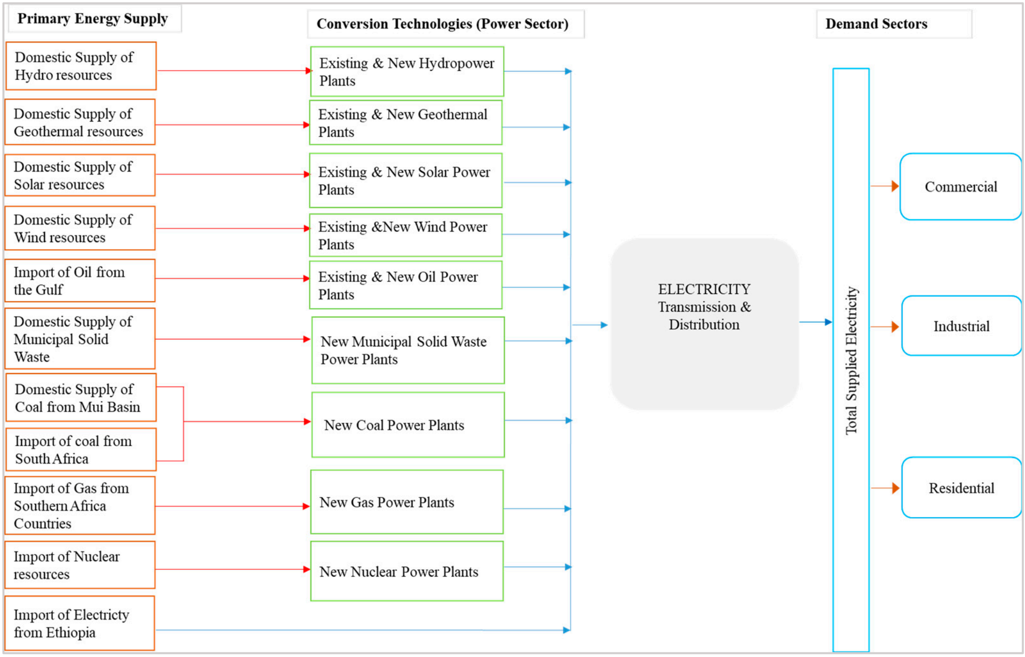

Figure 5. The reference energy system (modified from Musonye et al. [15]). To the left are the domestic and imported primary energy resources; in the middle are the conversion technologies used to generate electricity from the primary energy sources; there is also a box in the middle representing the transmission and distribution system that connects the generation with the demand sectors shown to the right.

Table 1. The available and developed domestic primary energy resources potential as used in the model.

Resource	Available Potential (MW)	Developed Potential (MW)
Geothermal	4000	865
Hydropower	2300	826
Wind	4600	336
Solar	70,000	50
Coal	95,000	0
Natural Gas	4000	0
Biomass	3500	0

3.2.2. Conversion Technologies

The conversion technologies considered are electricity-generating plants, including hydro, oil, geothermal, biomass, wind, solar, and nuclear power plants (see Figure 5). The technical parameters (see Table 2) at the start of the modelling horizon are adopted from the government's power planning LCPDP reports [22,45]. Technology learning curves are considered for all the technologies, beginning with the base year's costs and projected future costs based on cost reduction rates adopted from the IEA's technology database [49]. The capacity factors and efficiency for the generation technologies existing in the base year are computed from the base year's installed capacity and the energy generated in the same year. On the other hand, coal, gas, and nuclear capacity factors are adopted from the LCPDP report [45]. The biomass technical parameters and fossil fuel GHG emission coefficients were adopted from the IEA's database [49], while renewable energy lifecycle emission assessments (LCAs) are from different research papers.

Table 2. Technical parameters for the conversion technologies considered in the model. CC is capital costs, FOM is fixed operation and maintenance costs, and VOM is variable operation and maintenance costs.

Conversion Technologies	CC USD Million/GW	FOM USD Million/GW	VOM USD Million/PJ	Capacity Factors	Contribution to Peak (%)
Geo	3595	155.4	1.0	0.90	95
Hydro	3741	19.6	0.5	0.65	95
Oil	2032	32.0	10.0	0.20	95
Solar	1780	27.3	0.0	0.19	0
Gas	1083	25.2	13.5	0.75	95
Nuclear	8068	7.5	10.0	0.85	95
Wind	2132	76.0	0.0	0.36	20
Biomass	3045	50.2	9.0	0.80	90
Coal	2542	75.4	(domestic) 3.5 (imported) 4.3	0.85	95

3.2.3. Demand Sectors

Since the total demand is exogenously input, demand in the reference energy system is simplified to residential, commercial, and industrial, including transmission and distribution losses. The parameters used to project future demand are adopted from the government's power planning reports [22,45] and are projected to increase by 8% annually. The peak reserve capacity is 20% [22].

3.2.4. Other Model Parameters

The Kenya-TIMES model base year was set in 2020, and the modelling horizon is from 2020 to 2050 with five-year investment time periods (t). The general discounting factor is 7% [50], and the energy unit for the transformation processes is a petajoule (PJ) [46].

3.2.5. Model Assumptions

Some of the model's general assumptions are that there will be no financial insufficiencies since the private sector is expected to be highly involved in future power sector development [51]. Hence, no constraint is imposed on the availability of financial resources, which may not fully represent reality in the governmental or private sectors. Power export and off-grid generation are also not considered in the current model.

3.3. Scenario Development

The study uses the BAU scenario as a reference case, which evaluates four additional scenarios based on Kenya's demand–supply expansion plan [22]: the Kenya National Electrification Strategy [52], Kenya's Energy Act 2019 [29], the Kenya National Energy Policy [53], the Import Tax Act concerning solar and wind power development accessories [54] and the environmental policy goals outlined in the NDC for Kenya. These scenarios focus on evaluating policy instruments for stimulating low-carbon power generation.

The BAU scenario assumes that there are no new policy interventions.

In the CTax scenario (Carbon Tax), a GHG emission tax is levied on all GHG emissions associated with power generation. A carbon tax has been used in many countries as a tool for reducing GHG emissions [55,56]. Based on the analysis done for Africa's power sector [57] and projected global carbon pricing [32,58], the carbon tax scenario was set at

USD 10/ton and USD 50/ton in 2022 and 2050, respectively, and the model interpolates the costs between 2022 and 2050.

The RPS scenario (Renewable Portfolio Standard) is a renewable energy target generation scenario in which a particular share of renewable energy-based power generation is implemented. Although the Kenyan government aims to supply all its citizens with clean and affordable energy, it does not have a defined targeted share of renewable energy generation beyond 2022, after which it plans to meet the increased electricity demand using renewable energy and fossil fuel sources. The study defines the renewable energy share in 2050 to be 100%, pegged on Kenya's abundant renewable energy resources potential. It should be noted that the value is not based on empirical studies or published policy documents.

In the RenSub scenario (Renewable Energy Subsidy), a capital subsidy is defined for all the new solar photovoltaic and wind capacities. With the recent scrapping of the 14% VAT exemption for imported specialised solar and wind equipment by the Kenyan government [54], assessing other alternative policy interventions is essential. Solar photovoltaic and wind technologies were considered for subsidies because the resources are fairly distributed in the country and can be installed in smaller units than other renewable energy technologies. Moreover, the recently enacted net metering regulation [29] enables individual home solar systems and small wind turbines to supply power to the primary grid. The subsidy value is 18% based on the price-gap method [59] and the grid-connected feed-in tariff Kenyan policy document [60].

Inspired by actual examples, the TaxSub scenario (Tax Subsidy) tests the concurrent implementation of the carbon tax and renewable energy subsidy policy instruments [61]. The scenario assumes that the proceeds from the CTax scenario are used to subsidise wind and solar power development.

4. Results

The following subsections describe the model results for the five scenarios based on the installed capacity, power generation, GHG emissions, and system costs.

4.1. Installed Capacity

The Kenya-TIMES results show notable energy transitions in the low-carbon scenarios where the share of the capacity of renewable energy sources, mainly solar, increases, and the share of coal capacity decreases compared with the BAU.

In Carbon Tax (CTax) scenarios, the share of the capacity of solar power increases rapidly, reaching 34% by 2050, as the economic feasibility of coal drops due to the rising carbon tax. The Renewable Portfolio Standard (RPS) and Tax Subsidy (TaxSub) scenarios introduce solar earlier, with the TaxSub initiating solar adoption in 2025. By 2050, solar will reach a 59% share of capacity in the RPS scenario and 50% in the TaxSub due to earlier adoption and aggressive support through the combination of the carbon tax and the subsidies. Conversely, in the Renewable Energy Subsidy (RenSub) scenario, without a carbon tax, coal and gas remain competitive, which limits the use of biomass to replace coal.

The total installed capacity varies across the scenarios, as seen in Figure 6. The TaxSub scenario has the highest installed capacity at 50 GW, while the BAU has the lowest at 24 GW. The high level of installed capacity is due to the share of intermittent sources, particularly solar, in the capacity mix. Solar radiation is only available between 7 am and 6 pm throughout the year in Kenya and has an overall capacity factor of 20% (see Kenya's daily solar radiation profile and average daily load curve in Figure 4). Therefore, the model installs higher solar capacity to meet demand and account for the low capacity factor.

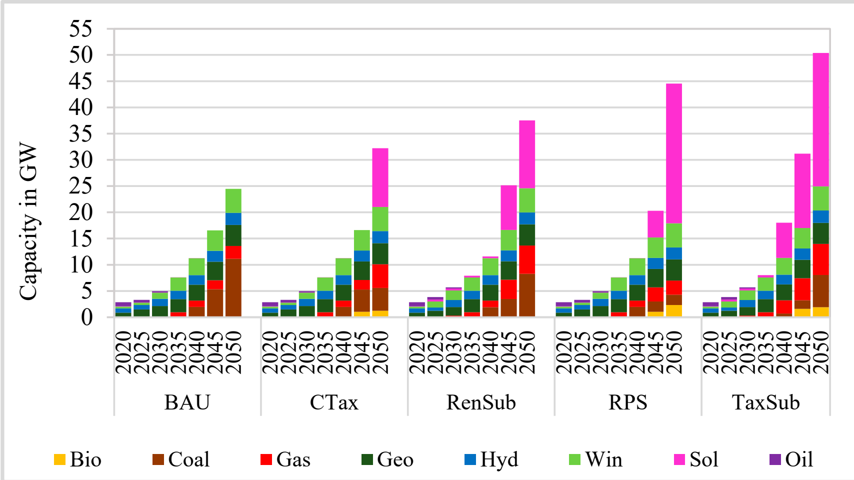

Figure 6. Installed power capacity across the five scenarios from 2020 to 2050. Low-carbon scenarios show a significant shift toward renewable energy, with solar capacity increasing substantially, particularly in the RPS and TaxSub scenarios. The TaxSub scenario has the highest installed capacity, driven by the need to compensate for the low solar capacity factor, while the BAU has the lowest.

4.2. Power Generation

The electricity generation profiles are similar to the installed capacity in the corresponding scenarios (see Figure 7). For instance, in 2050, coal will generate 56% of the total energy in the BAU scenario, while it will generate 0% in the RPS scenario in the same year. In other low-carbon scenarios, coal generation is partly replaced with solar and biomass generation. Domestic coal supply is prioritised in coal power generation because of its low costs, and the high costs associated with oil-based generation prohibits its use. In all the scenarios, imported power becomes competitive beginning in 2035 after the exhaustion of geothermal, hydropower, and wind resources, with the system consuming the maximum available capacity. The 100% generation from renewable constraints in the RPS scenario will replace coal with biomass by 2050.

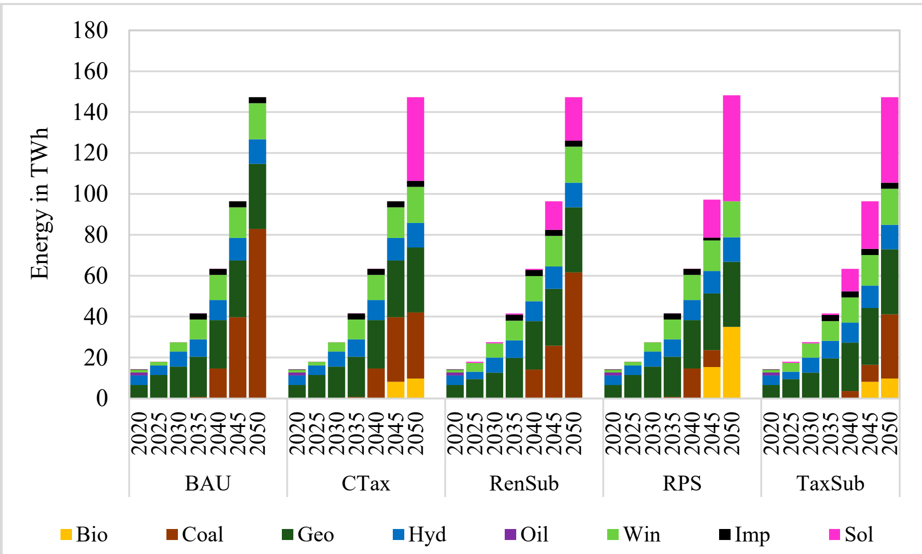

Figure 7. Electricity generation mix across the five scenarios from the year 2020 to 2050. Power generation trends follow installed capacity patterns, with coal supplying 56% of total energy in the BAU by 2050 but being completely phased out in the RPS scenario. In low-carbon scenarios, coal generation is largely replaced by solar and biomass, with imported power becoming competitive from 2035 onward as geothermal, hydropower, and wind resources are exhausted.

Interestingly, despite the installed gas capacity, no energy is generated from the gas units. This is due to the peaking reserve constraint in the model, which specifies that the available capacity must exceed the demand of a selected commodity by a certain margin in each time slice. Except for the RPS scenario, which has a strict constraint on the share of energy that should come from renewables, the natural gas installed capacity increases with the intermittent wind and solar capacity. As indicated in Figure 8, the higher the percentage of power generated with intermittent sources, the more peaking reserve is required, and the model selects to install gas capacity due to low capital costs and low emissions if not used to generate power.

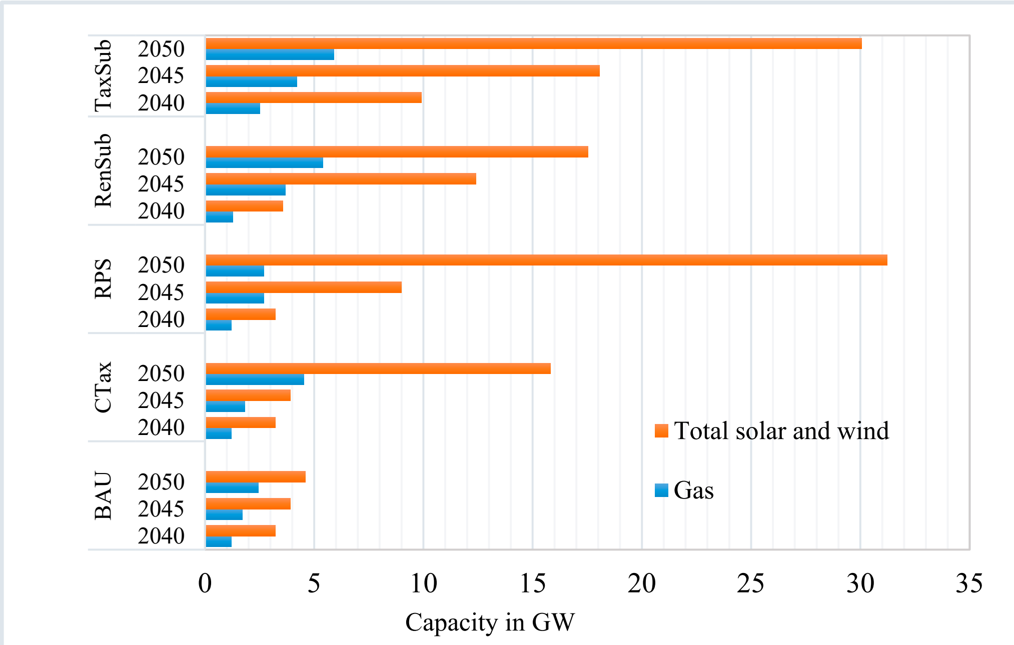

Figure 8. A chart showing the relationship between the total solar, wind, and gas capacities in the last fifteen years of the modelling period. The natural gas installed capacity for peak reserve increases with the intermittent wind and solar capacity.

4.3. GHG Emissions

Figure 9 presents the emissions under the five scenarios compared with the government's projected power sector-related GHG baseline emissions and the NDC's allowable emission level. The emissions in all five scenarios are predominantly below the NDC's level until 2045.

The government's power sector-related GHG baseline emission projections are based on power generation expansion simulated using an econometric model. Unlike the current optimisation study, the government's power generation forecast envisages the use of fossil fuels earlier in the modelling period, and coal and natural gas generation is expected to come on the grid in 2025 and 2028, respectively [22,45]. The sharp increase in emissions after 2035 relates to the uptake of coal generation in 2040 in all the scenarios. After 2045, the emissions in the RenSub scenario are above the NDC permitted level by 12%, mainly due to the very sharp increase in the use of coals. Due to the high share of coals, the BAU emissions are 4% above the government's projected baseline and 49% above the NDC allowed emission levels. The RPS scenario is bound to have insignificant emissions in 2050 because of the 100% renewable generation constraint. The growing carbon tax on coal in the CTax scenario results in reduced emissions as coal generation will be partly replaced by solar energy in the CTax scenario after 2045. The slightly high emission level in the base year relates to the diesel generators, whose contribution to the energy mix ends in the base year.

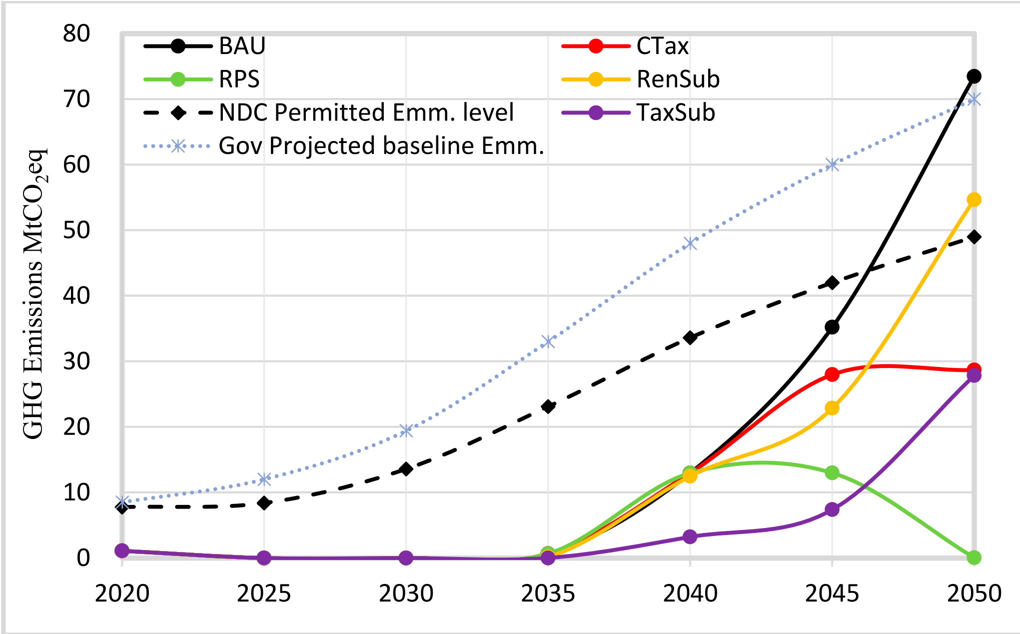

Figure 9. GHG emissions for the five scenarios projected baseline emissions and the acceptable level of emissions under NDC. Both the BAU and the RenSub scenarios exceed NDC levels due to the steep increase in the use of coal resources.

4.4. System Costs

A comparison of the total system costs, as defined by Equation (1), is presented in Figure 10. The RPS scenario's total system costs are the highest of the five scenarios being 33% higher than the RenSub, which has the lowest costs. The high share of solar in the RPS scenario accounts for the high system costs. In contrast, the 18% subsidy on investment in wind and solar capacity, catered for by the taxpayers, accounts for the lowest costs, as expressed in the RenSub scenario. For the same reason, the subsidy scenarios have lower costs than the BAU.

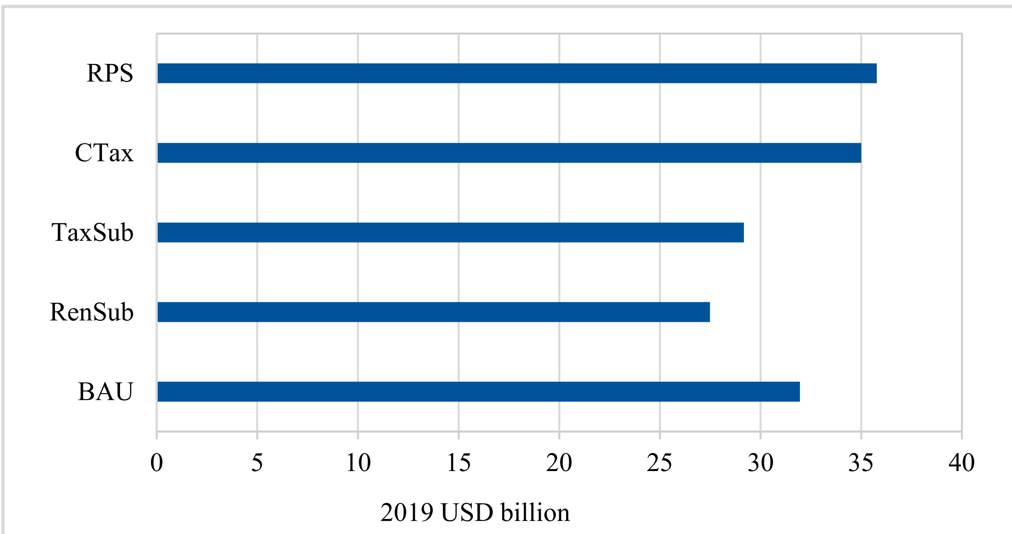

Figure 10. Total discounted system costs for the five scenarios in 2019 in USD billion.

The CTax scenario's total system costs are 9% more than the BAU. With the exhaustion of the most economical geothermal, hydropower, and wind resources potential, the CTax scenario remains with the taxed coal and the expensive solar generation options.

The breakdown of the total system costs in Figure 11 account for the non-discounted annual investment (INV), fixed operation and maintenance (FIX), and variable investment (VAR) costs and compares the costs incurred in the same years under different scenarios. The base year's high VAR costs resulted from the fuel costs required to run the diesel generators, which generated 10% of the energy in that year; however, these diesel generators were not used after 2020. Due to the higher installed solar capacity, the low-carbon development scenarios have higher INV and FIX costs but lower VAR costs than the BAU scenario from 2045.

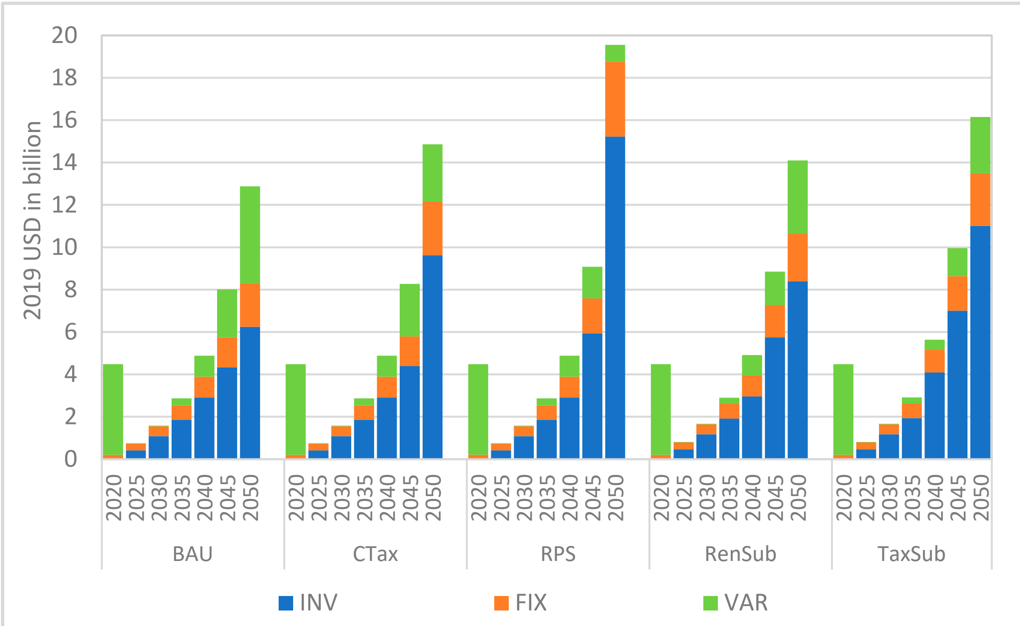

Figure 11. Total non-discounted system costs for the five scenarios in 2019 in USD billion.

The unit cost of electricity in the current study is presented using the marginal electricity price at the distribution point. The marginal electricity price is the cost incurred to generate an extra unit of electricity and includes the capital costs, fixed costs, and variable costs [62]. There is a general drop in marginal prices after 2020 across the scenarios, as shown in Figure 12, due to the removal of diesel generators. The subsidy component accounts for the more significant price drop in the RenSub and TaxSub scenarios. The general increase in marginal prices after 2030 in the BAU, CTax, and TaxSub scenarios is accounted for by coal capacity, which comes onto the grid in 2035. The carbon tax element in the CTax and TaxSub scenarios explains the further price increase compared with the BAU. Combined with the lower investment and fixed costs resulting from the subsidy, the TaxSub marginal prices are slightly below the CTax scenario. The marginal prices in the TaxSub scenario are higher than the BAU because of the carbon tax levied on GHG emissions resulting from generated power. The rise in marginal prices in the RPS scenario from 2040 onwards relates to the increased share of solar power. The RenSub has the lowest marginal prices among the five scenarios due to the externally funded subsidy.

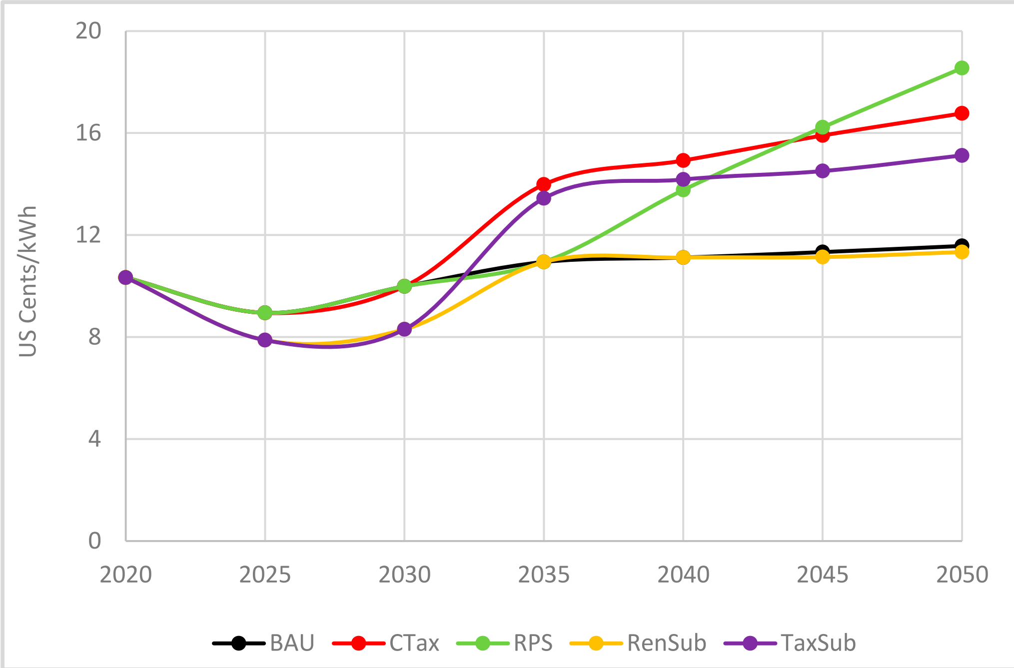

Figure 12. Marginal electricity prices at the distribution point for the five scenarios.

5. Discussion

Sub-Saharan Africa (SSA), a region with around 14% of the global population, is at significant risk from climate change, which threatens its ecosystems and economies, as reported by the IPCC [63]. Concurrently, SSA has some of the world's lowest rates of access to electricity and clean cooking energy, with traditional biomass constituting nearly half of the primary energy mix. Hence, curbing GHG emissions while enhancing access to affordable, clean energy is crucial for the region [7]. In response to climate change and global warming, governments and energy stakeholders worldwide strive to lower power sector-related GHG emissions by crafting and implementing strategies for low-carbon power generation pathways, utilising energy policies assessed and enacted through various policy instruments [64]. Kenya exemplifies a Sub-Saharan African country poised for a substantial rise in power demand in the coming decades. Despite its abundant renewable energy resources, Kenya lacks a strategy to guide the expansion of low-carbon power generation.

The study in this paper proposes the Kenya-TIMES model to evaluate how low-carbon strategies affect Kenya's power generation expansion. The model's structural improvements include a time slice structure for a detailed representation of the capacity factors of intermittent renewables, specifically solar and wind. While not explored in this study, the enhanced structure can also be used to examine the potential for energy storage systems. This model, grounded in the best actual data, incorporates all viable energy sources and conversion technologies within the country. Given the complexity of energy system development, such a tool is essential for devising optimal long-term energy strategies.

The current study underlines that the cost-competitive nature of geothermal, hydropower, and wind presents these renewables as prime resources for power generation expansion in Kenya. Due to the predicted steep increase in energy demand, the available potential of these resources will be fully used in 2035. After that, the low-carbon strategies will significantly impact the continued expansion of the power generation technology mix. The model also underlines the strong effect of the peaking reserve constraint on the amount of installed capacity when intermittent renewable energy sources are prioritised in the generation mix. In 2050, for instance, the TaxSub scenario has 25 GW of installed solar capacity compared with zero in the BAU and 13 GW in the RenSub. The TaxSub has a 1.9 GW biomass capacity in 2050 compared with zero in the BAU and RenSub in the same year to counterbalance the high level of intermittent solar capacity. Furthermore, the hydropower capacity is 17.65% higher in the TaxSub than in the BAU and RenSub in 2050, while the gas capacity is 61% and 8.8% more than the BAU and RenSub scenarios, respectively, in 2050. The installed gas capacity is not used for power generation to limit emissions; it is just a reserve. These dynamics show the role that a renewable energy subsidy or a hybrid of the renewable energy subsidy and the carbon tax scenarios could play in expediting the development of wind and solar resources in Kenya's power sector in the short to medium term.

The energy generation in the RPS scenario, like the BAU, is 100% renewable from the base year until 2035. After that, the effects of the RPS targets are seen most significantly after 2045, as the energy system aims to attain 100% renewable energy generation in 2050. Unlike in the TaxSub and RenSub scenarios, the increase in solar capacity is not commensurate with the accelerated gas capacity development in the RPS case. In the RPS scenario, the last installed capacity of coal is developed in 2040, with an economic lifetime of 30 years, yet, in 2050, there is no power generation expected from the coal plants, which results in 1.9 GW of idle capacity. The idle coal capacity is counted as reserve capacity but could not, in fact, be used due to the RPS 100% renewable energy target. With this RPS 100% renewable generation requirement—but without storage—the gas plants would not be sufficient to balance the intermittency associated with solar and wind, and the system would be impractical in real-life situations. In addition to the RPS scenario in 2050, the TaxSub has the lowest coal generation among the low-carbon scenarios in the 2040–2050 period. These low capacities indicate that a carbon tax and subsidy hybrid could be more effective in reducing the share of fossil fuels in Kenya's power generation mix in the long term.

The GHG emission reduction under the low-carbon scenarios indicates that policy instruments can be used to regulate Kenya's power sector-related GHG emissions and achieve emission reduction targets. High GHG emission cuts in the low-carbon scenarios correlate with higher solar and biomass resource development instead of coal. This replacement indicates the central role of solar and biomass resources in decarbonising Kenya's future power sector. In all scenarios, the model results provide emission totals from 2020 to 2035 that are 87% below the government's NDC projection. The government's generation forecast assumes a significant share of fossil fuel generation as early as 2020. However, in the current study, geothermal, hydropower, and wind power generation dominate the 2020–2035 period. The low emissions suggest that the Kenyan government can avert coal development and the large-scale absorption of the expensive solar and biomass resources if it harnesses geothermal and hydropower to their full sustainable potential. Different studies have set Kenya's geothermal potential to be between 7 and 10 GW and wind potential between 4 and 6 GW. In contrast, the current study uses a conservative figure of 4 GW for geothermal and 4.6 GW for wind. Increasing the potential values to the upper limit would significantly affect the energy mix and resulting emissions.

The energy system costs are higher in the RPS and CTax scenarios, mainly because of the large installed capacities of the relatively expensive solar resources. However, the cost prediction of solar technology is not reliable this far ahead in time, and the rapid development of solar technologies may significantly decrease the costs. Furthermore, deploying storage technologies could substantially alter the system costs in the RPS and CTax scenarios. On the other hand, the RenSub and TaxSub scenarios show lower energy system costs than the BAU. However, the government shoulders the costs of these subsidies, resulting in a USD 5 billion revenue loss for the entire modelling period. An evaluation of the ability of the government budget to absorb the revenue losses would provide insight into the feasibility of this policy instrument on the government's side. Alternatively, the government can let the subsidy costs be financed by a subsidy surcharge, which would be added to the final electricity tariff, as has been the case in Spain and Germany [65]. In all the scenarios, marginal electricity price trends relate similarly to the total system costs except in the TaxSub scenario. The TaxSub assumes that the carbon tax levied on unit production is used to subsidise additional wind or solar units. This design removes the direct subsidy burden from the government's shoulders and discourages fossil fuel consumption while providing an option for renewable capacity development. This observation relates to applying such policy instruments in Germany, Britain, and Spain [61,65]. The introduction of storage could substantially alter the marginal prices associated with the TaxSub scenario.

Based on the assumptions in the current study, the overall analysis indicates that a hybrid of the renewable energy subsidy and the carbon tax in the TaxSub scenario obtains a feasible low-carbon development pathway for Kenya's future power development with emissions below the NDC allowable levels. The TaxSub attains solar adoption and accelerated uptake of wind resources earlier in the modelling horizon. Furthermore, the TaxSub has the second-lowest total system costs and the third-lowest marginal electricity prices and protects the government from revenue losses if the accrued subsidy costs do not exceed the levied carbon tax. Finally, it should be emphasised that the accumulated emissions over the entire modelling period are the lowest for the TaxSub scenario, which is even more important than the targeted point values at specific years.

The new Kenya-TIMES model structure and assumption refinement have improved the current results compared with the earlier Kenya-TIMES model results [15] and the government's LCPDP econometric model results [22]. The detailed representation of the time slices and wind capacity factor increased the competitiveness of wind resources to the same level as geothermal and hydropower. The competitiveness of wind resources and the inclusion of biomass options in the current model altered the adoption of nuclear capacity, which the other two models adopted. This observation indicates that a more advanced energy planning tool will likely inform effective and efficient decisions for sustainable energy development.

This study has some inherent limitations, and the current Kenya-TIMES model could be further improved to account for storage and the short-term operational constraints on Kenya's power generation expansion. The intermittent energy sources have fluctuating generation outputs, some with predictable short-term patterns (such as solar). In contrast, the output variation from others is more challenging to balance with energy storage solutions. Such accounting is critical for short- to medium-term planning. The analysis does not consider private or governmental financial limitations, which does not accurately reflect reality. Also, there is a need for a multi-criteria analysis to rank the low-carbon policy instruments based on energy security, reliability, costs, accrued revenue loss and gain by the government, affordability, and environmental benignity. Such analysis will provide proper insights into the most effective and efficient policy instrument for Kenya's sustainable energy development path. Finally, conducting an extensive sensitivity analysis on critical parameters, such as resource potential, technology costs, and their future trajectories, would strengthen the insights provided by this study. The modelling platform developed here facilitates such an analysis, offering policymakers a deeper understanding of the underlying dynamics and their implications for energy planning.

6. Conclusions

Kenya is an example of a Sub-Saharan African country with an expected rise in power demand in the following decades. Despite its abundant renewable energy resources, Kenya lacks a strategy and proper tools to guide the expansion of its low-carbon power generation. This study developed the Kenya-TIMES, an optimisation-based, national-scale, bottom-up energy system planning model, in order to assess the impact of low-carbon strategies on Kenya's power generation expansion. The model uses local energy data to incorporate advanced TIMES framework features, such as multiple time slices, technology learning curves, peaking reserve constraints, and specific capacity factors for intermittent energy sources.

The model results indicate that low-carbon policy instruments can reduce GHG emissions below Kenya's Nationally Determined Contributions (NDC) target by 2050. Even though the renewable portfolio standard achieves the highest emission cuts by 2050, it does so at the highest cost, and, without storage, it is infeasible in 2050. A combination of taxes and subsidies could provide an effective and efficient low-carbon pathway for Kenya's power generation expansion. Additionally, prioritising the development of geothermal and hydropower to their full potential could lead to significant emissions reductions at lower costs.

This study further underlines the need for sophisticated energy planning tools to make informed energy planning decisions and guide the selection of the most effective and efficient policy instruments for developing Kenya's sustainable energy path. To enhance its use for short- to medium-term planning, the Kenya-TIMES model could be further refined to include storage considerations and short-term operational constraints affecting Kenya's power generation expansion. Additionally, a comprehensive sensitivity analysis of key model parameters and assumptions would provide deeper insights into the uncertainties affecting Kenya's energy transition and strengthen the robustness of policy recommendations.

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